MATH 598 (Topics in Probability & Statistics).
MATH 782 (Advanced Topics in Statistics).
Course Catalogue –FALL 2026

Instructor: Archer Yi Yang
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Lecture: Tu, Thur 2:35 pm – 3:55 pm Burnside 920
Office Hours: Tu, Thur 4:00 pm – 5:00 pm
Prerequisites: MATH 356 Probability or equivalent. Familiarity with R/Python is recommended.

References: Optional textbook:

- Theoretical Foundations of Conformal Prediction by Anastasios N. Angelopoulos, Rina Foygel Barber, Stephen Bates

Links to other learning resources will be provided on MyCourses.

Course objectives This course provides a formal introduction to the conformal prediction framework, a model-agnostic uncertainty quantification method applicable to machine learning and artificial intelligence systems. The curriculum examines the mathematical mechanisms of distribution-free inference and their computational implementation. Students will derive finite-sample, model-agnostic coverage bounds utilizing exchangeability and permutation tests. The course connects theoretical foundations with applied methodology, analyzing the use of conformal inference in high-dimensional settings, trustworthy AI pipelines, computational biology, and drug discovery.

Audience Graduate students and advanced undergraduates in statistics, applied mathematics, biostatistics, computer science, and related fields.

Tentative course content

- Module I: Theoretical Foundations
  - Exchangeability and permutation tests.
  - Full and split conformal prediction algorithms.
  - Limits of distribution-free regression and the Bahadur-Savage theorem.
- Module II: Methodological Core and Score Design
  - Conformal score functions for regression and classification.
  - Limits of test-conditional coverage in continuous settings.
  - Relaxations of conditional coverage.
  - Asymptotic convergence guarantees.
- Module III: Algorithmic Extensions and Complex Data
  - Handling distribution shifts via likelihood ratio reweighting.

  - Cross-validation methods: cross-conformal, CV+, and Jackknife+.
  - Online conformal prediction and conformal risk control.
  - False discovery rate (FDR) control in outlier detection.

- Module IV: Applied Case Studies in CS and Biostatistics
  - Conformal risk control for computer vision outputs.
  - Uncertainty quantification in drug discovery and high-dimensional biological data.
  - Prediction sets for large language models and AI pipelines.

Grading

- Homework assignments (30%): Regular problem sets covering mathematical derivations and computational implementations. In a team of two, or individually.
- Presentation (20%): Analysis of an advanced methodological paper or domain-specific conformal inference pipeline, detailing the statistical theory and empirical outcomes. In a team of two, or individually.
- Team project (40%): A collaborative research project consisting of a final written report (20%) and a class presentation (20%). Teams are limited to two students. The final report must describe the individual contributions of each member.
- Class participation (10%): Assessed via engagement in lecture discussions.
- Exam: none

Policy on collaboration: Collaboration on homework assignments with fellow students is encouraged. However, such collaboration should be clearly acknowledged, by listing the names of the students with whom you have had discussions concerning your solution.

Feedback: I welcome comments and suggestions (and complaints) and will solicit feedback via a survey partway through the class. Comments at any other time are welcome, and if you prefer anonymity, you can leave a note in my mailbox or under my door.

When to see me about an assignment: I'm here to help, including providing guidance on assignments. But before coming to see me about a difficulty, you should try something a few different ways and try to define/summarize what is going wrong or where you are getting stuck.

+ **In the event of extraordinary circumstances beyond the University's control, the content and/or evaluation scheme in this course is subject to change. If you need special assessment arrangements or accommodations, please contact the Office for Students with Disabilities at 514–398–6009.**

Assessments and Plagiarism: Any material (assignment solution scripts, R scripts etc.) that you hand in for assessment should be your work alone, unless the assignment concerned is a designated group project. This does not mean that you cannot consult or collaborate with others, merely that the material submitted for assessment must have been produced by you.

Shortlist of Resources: The Department of Mathematics and Statistics, as well as the University at large, have many resources to help you succeed in this course and throughout your degree. A more extensive list of resources can be found in the MyCourses Webpage for this Course (under "Content"), as well as the Department's EOSW webpage. A shortlist of the most commonly used resources is available here:

- The Wellness Hub is a centralized website for student physical and mental health resources.
- The Math Help Desk is staffed by knowledgeable math students who can help answer your questions related to your courses. They have tutors on M-F from noon-5 PM in Burnside Hall room 911.
- Statistics Online for Students is a resource which offers online help on statistics courses in the department. Click on the link above and you will be added to a Microsoft Teams group where the schedule will be announced.
- SUMS is the Society of Undergraduate Mathematics Students. Join their Facebook group to get on their listserv, connect with other students in the department, and participate in some of their activities/social events.
- GSAMS is the Graduate Student Association for Mathematics and Statistics (GSAMS). All graduate students are represented by GSAMS, and they hold regular events throughout the semester to promote graduate student community.
- Advising is an important resource to guide you throughout the duration of your degree, and advisors can help with answering questions related to your degree program. Check out the Department's Advising Website to find out how to get in touch with a Departmental Advisor, should you need to do so.
- The Office for Mediation and Reporting is a McGill centralized office used to file a formal report of discrimination, harassment, or sexual violence; learn about policies and processes; or be connected to additional supports.
- The Office for Sexual Violence, Response, Support and Education provides support for all members of the McGill community who have been impacted by sexual violence (whether it be sexual harassment or assault, gender-based or intimate partner violence, or cyberviolence) and works to foster a culture of consent on campus and beyond.
- The Office of the Dean of Students (Case Management) is a collaborative process between a student, a Case Manager, and other concerned parties with the intention of improving the student's academic and personal outcomes. Case managers are trained in confidentiality, disclosures of sexual violence, and mental health first aid.

MCGILL UNIVERSITY POLICY STATEMENTS

The following three statements are included in this course outline, in keeping with Senate resolutions:

1. *McGill University values academic integrity. Therefore, all students must understand the meaning and consequences of cheating, plagiarism and other academic offences under the* ***Code of Student Conduct and Disciplinary Procedures****. For more information, see*

   `www.mcgill.ca/students/srr/honest/`

   [*Approved by Senate on 29 January 2003*]

2. *In accord with McGill University's Charter of Students' Rights, students in this course have the right to submit in English or in French any written work that is to be graded.*

   [*Approved by Senate on 21 January 2009*]

3. *Instructors who may adopt the use of text-matching software to verify the originality of students' written course work must register for use of the software with Educational Technologies and must inform their students before the drop/add deadline, in writing, of the use of text-matching software in a course.*

   [*Approved by Senate on 1 December 2004*]

If you have a disability and need special arrangements, please contact the **Office for Students with Disabilities** at 514–398–6009.